

University AI Database Chatbot

Project Overview

The University AI Database Chatbot is an AI powered assistant that allows users to interact with a university database using natural language.

Instead of writing SQL queries manually, users can ask questions in plain English, and the system automatically converts them into SQL queries, retrieves data from the database, and returns the results.

Example:

User: "What is the salary of Dr. Mehta?"

Generated SQL:

```
SELECT salary  
FROM faculty  
WHERE name='Dr. Mehta';
```

Output:

Dr. Mehta's salary is ₹120000.

Features

- Natural Language Querying
 - AI Powered SQL Generation
 - SQLite Database Integration
 - Faculty Management
 - Student Management
 - Course Management
 - Attendance Tracking
 - Marks Management
 - Administration Records
 - Dynamic Query Handling
 - GitHub Version Control
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Technologies Used

Programming Language

- Python

Database

- SQLite

AI Model

- Groq LLM (Llama 3)

Libraries

- sqlite3
- groq
- os

Version Control

- Git
 - GitHub
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Database Schema

Students Table

Column	Description
student_id	Unique student ID
name	Student name
age	Student age
gender	Student gender
department	Department
semester	Current semester

Faculty Table

Column	Description
faculty_id	Faculty ID

Column	Description
name	Faculty name
department	Department
designation	Job title
salary	Faculty salary

Courses Table

Column	Description
course_id	Course ID
course_name	Course name
faculty_id	Faculty assigned

Marks Table

Column	Description
mark_id	Record ID
student_id	Student reference
course_id	Course reference
marks	Marks obtained

Attendance Table

Column	Description
attendance_id	Attendance ID
student_id	Student reference
course_id	Course reference
attendance_percentage	Attendance percentage

Administration Table

Column	Description
admin_id	Admin ID
name	Admin name

Column	Description
role	Job role
salary	Salary

System Architecture

User Question
↓
Groq LLM
↓
SQL Query Generation
↓
SQLite Database
↓
Query Execution
↓
Result Retrieval
↓
Answer Display

Working Process

Step 1

User enters a question.

Example:

"How many students are in CSE?"

Step 2

The question is sent to Groq LLM.

Step 3

The LLM studies the database schema.

Step 4

The LLM generates a valid SQL query.

Example:

```
SELECT COUNT(*)  
FROM students  
WHERE department='CSE';
```

Step 5

Python executes the SQL query using SQLite.

Step 6

Database returns the result.

Step 7

The result is displayed to the user.

Sample Questions

- Show all students.
 - Show all faculty members.
 - What is the salary of Dr. Mehta?
 - How many students are in CSE?
 - Who teaches Data Structures?
 - Show attendance below 80%.
 - Which student scored highest marks?
 - List all administration staff.
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Future Enhancements

- Streamlit Web Interface
 - Authentication System
 - Role-Based Access Control
 - Voice Assistant Support
 - RAG Integration
 - PDF Report Generation
 - Multi-University Support
 - Fine-Tuned Domain Models
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Author

Sanat Kumar Mishra

AI Powered University Database Chatbot Project